

- capacitive load 540
 - capture clock 172, 173, 174, 367, 370
 - capture clock edge 326
 - capture flip-flop 36
 - CCB 80
 - CCS 47, 76
 - CCS noise 80
 - CCS noise models 85
 - CCSN 80
 - ccsn_first_stage 82, 84, 85, 86
 - ccsn_last_stage 84, 85, 86
 - cell check delays 369
 - cell delay 368, 467, 469
 - cell instance 473
 - cell library 12, 113, 153, 392
 - cell placement 547
 - cell_rise 52
 - channel connected blocks 80
 - channel length 366
 - characterization 54
 - check event 482
 - circuit simulation 532
 - clock cycle 318
 - clock definitions 2
 - clock domain 36, 273, 435
 - clock domain crossing 10, 445
 - clock gating 365, 406, 413
 - clock gating check 192, 394
 - clock latency 30, 188
 - clock period jitter 31
 - clock reconvergence pessimism 373
 - clock reconvergence pessimism removal 370
 - clock skew 30
 - clock source 181
 - clock specification 181
 - clock synchronizer 10, 38, 445
 - clock tree 6, 236, 370
 - clock tree synthesis 189
 - clock uncertainty 186, 335
 - clock_gating_default 399
 - closing edge 377
 - CMOS 5
 - CMOS gate 16
 - CMOS inverter 16
 - CMOS technology 15
 - combinational cell 33
 - comment 538
 - common base period 306
 - common clock path 370, 375
 - common path pessimism 370, 375
 - common path pessimism removal 370
 - common point 370
 - composite current source 76
 - COND 72
 - conditional check 510
 - conditional hold time 510
 - conditional path delay 477, 479
 - conditional propagation delay 502
 - conditional recovery time 511
 - conditional removal time 513
 - conditional setup time 508
 - conditional timing check 482
 - connection attribute 540, 543
 - connectivity section 543
 - constrained pin 385, 392
 - constrained_pin 63
 - controlled current source 115
 - coordinates 540
 - coupled nets 118
 - coupling capacitance 118, 149, 544
 - CPP 370
 - CPPR 370
 - create_clock 182, 453
 - create_generated_clock 190, 454
 - create_voltage_area 466
 - critical nets 120
 - critical path 6, 247
 - cross-coupling capacitance 542, 543
 - crosstalk 2, 121
 - crosstalk analysis 147, 532
 - crosstalk delta delay 149
 - crosstalk glitch 83, 160
 - crosstalk noise 147, 163
 - CRPR 370
 - current loops 102
 - current spikes 148
 - current_design 450
 - current_instance 448
 - cycle stealing 377
- D**
- D_NET 532, 542
 - DAC interface 360
 - data to data check 385
 - data to data hold check 385
 - data to data setup check 385
 - DC margin 87, 154
 - DC noise analysis 156
 - DC noise limits 153
 - DC noise margin 153, 157
 - DC transfer characteristics 153

dc_current 82, 153
DDR xix
DDR interface 121
DDR memory 10
DDR SDRAM 317
DDR SDRAM interface 341
deep n-well 177
default conditional path delay 479
default path delay 477
default path group 209
default wireload model 112
define definition 534, 541
delay 474
delay specification 480
delay-locked loop 336
derate specification 374
derating 367
derating factor 96, 97, 368
design name 534
design rules 215
Design Under Analysis 180
detailed extraction 104
device delay 477, 519
device threshold 41
diffusion leakage 92
distributed delay 470
distributed net 532, 542
distributed RC 149, 545
distributed RC tree 103
distributed timing 477
DLL 336, 343, 349
DQ 341
DQS 341
DQS strobe 341
drive strength 211
driver pin 532, 548
driver reduction 548
driving cell 540
DSPF 113
DUA 3, 180, 317, 336
duty cycle 181

E

early path 35
ECSM 47, 76
edge times 181
EDIF 536
effective capacitance 75
effective current source model 76
electromigration 13
e-limit 476

Elmore delay 548
enclosed wireload mode 110
endpoint 207
entity instance 541
environmental conditions 96
error limit 476
escaped 550
escaped character 550
exchange format 531
expr 448
external definition 534, 540
external delay 206
external input delay 204
external load 536
external slew 536
extraction 119
extraction tool 7, 119
extrapolation slope 107

F

fall delay 51
fall glitch 152, 159
fall transition 51
fall_constraint 64
fall_glitch 154
fall_transition 51
false path 11, 38, 179, 272, 444
fanouts 21
fast clock domain 289
fast process 39, 96
file size 534
final route 7, 547
flip-flop 3
footer 414
forward-annotation 469, 471, 485
FPGA 5
frequency histogram 246
function specification 95
functional correlation 162
functional failures 5
functional mode 220

G

gate oxide tunneling 92
gating cell 394
gating pin 394
gating signal 394
generated clock 190, 328, 396, 435
generic 485
generic name 500
get_cells 450

INDEX

get_clocks 450
get_lib_cells 451
get_lib_pins 451
get_libs 451
get_nets 451
get_pins 451
get_ports 451
glitch 159, 470
glitch analysis 10, 147
glitch height 153
glitch magnitude 151, 153, 161
glitch propagation 159
glitch width 87, 153
global process variation 423
global route 7, 547
ground net 540
grounded capacitance 102, 118, 151, 164
group_path 455
guard ring 177

H

half-cycle path 274, 442
hardware description language 467
header 414
header definition 534
header section 471
hierarchical block 175, 472
hierarchical boundary 110, 542
hierarchical instance 473
hierarchical methodology 119
hierarchy delimiter 537
hierarchy separator 472, 473
high transition glitch 159
high Vt 92, 416
high-fanout nets 443
hold 62
hold check 3, 227
hold check arc 60
hold gating check 400
hold multicycle 262, 289
hold multiplier 269
hold time 509
hold timing check 248, 470, 474, 482, 510
hold_falling 393
hold_rising 393

I

ideal clock tree 30
ideal clocks 9

ideal interconnect 7, 9, 490
ideal waveform 25
IEEE Std 1076.4 499
IEEE Std 1364 496
IEEE Std 1481 531
IEEE Std 1497 468
IMD 431
inactive block 414
inductance 102, 547
inductance section 547
inductance unit 538
inertial delay 157
input arrival times 240
input constraints 319
input delay constraint 203
input external delay 255
input glitch 157
input specifications 206
input_threshold_pct_fall 25
input_threshold_pct_rise 25
insertion delay 188, 236
instance name 538
inter-clock uncertainty 187
interconnect capacitance 102
interconnect corner 418, 419
interconnect delay 467, 469, 477, 479
interconnect length 107
interconnect modeling 471
interconnect parasitics 101, 534
interconnect path delay 518
interconnect RC 419
interconnect resistance 102, 120, 419
interconnect trace 101, 102
inter-die device variation 423
inter-metal dielectric 431
internal definition 534, 542, 545
internal pin 543
internal power 88
internal switching power 88
intra-die device variation 424
IO buffer 43
IO constraints 218
IO interface 337
IO path delay 475, 477
IO timing 179
IR drop 366
is_needed 82

J

jitter 31

K

k_temp 99
k_volt 98
k-factors 96, 97

L

L value 534
label 474, 485
latch 377
late path 35
latency 440
launch clock 174, 370
launch edge 303
launch flip-flop 36
layout extracted parasitics 119
leakage 19, 415
leakage power 88, 92, 412, 416
Liberty 26, 43, 94
library cell 43
library hold time 253
library primitive 471
library removal time 279
library time units 100
linear delay model 46
linear extrapolation 107
list 448
load capacitance 46, 542
load pin 532
local process variation 424
logic optimization 5
logic synthesis 485
logic-0 19
logic-1 19
logical hierarchy 541
logical net 536, 545
logical port 540
longest path 34
lookup table 48, 64
low transition glitch 159
low Vt 93, 416
lumped capacitance 532, 548

M

master clock 190, 328
max capacitance 215
max constraint 229
max output delay 327
max path 34, 172
max path analysis 166
max path check 323
max timing path 229

max transition 215
max_transition 58
maximal leakage 41
maximum delay 482, 502
maximum skew timing check 470
metal etch 430
metal layers 101
metal thickness 431
Miller capacitances 82
Miller effect 76
miller_cap_fall 82
miller_cap_rise 82
min constraint 250
min output delay 327
min path 34, 172
min path analysis 166
min path check 323
minimum delay 482, 502
minimum period timing check 470
minimum pulse width timing
check 470
MMMC 421
MOS devices 92
MOS transistor 15
multi Vt cell 416
multicycle 444
multicycle hold 264
multicycle path 179, 260, 292
multicycle setup 264
multicycle specification 285, 335, 390
multi-mode multi-corner 421
multiple aggressors 160

N

name directory 119
name map 534, 538, 542
narrow glitch 155
negative bias 417
negative crosstalk delay 166, 167
negative fall delay 170
negative hold check 65
negative rise delay 170
negative slack 246
negative unate 33, 59
negative_unate 52
neighboring aggressors 149
neighboring signal 102
net 101
net delay 368, 479, 518
net index 542
net name 538

- netlist connectivity 536
- network latency 188
- NLDM 47, 75, 393
- NMOS 15
- NMOS device 414
- NMOS transistor 16
- no change timing check 471
- no-change data check 391
- no-change hold time 516
- no-change setup time 516
- no-change timing check 483
- no-change window 391
- noise 2, 83
- noise immunity 87
- noise immunity model 87
- noise rejection level 155
- noise tolerance 155
- noise_immunity_above_high 87, 159
- noise_immunity_below_low 87, 159
- noise_immunity_high 87, 159
- noise_immunity_low 87, 159
- nom_process 96
- nom_temperature 96
- nom_voltage 96
- nominal delay 502
- nominal temperature 41
- nominal voltage 41
- non-common 174
- Non-Linear Delay Model 47
- non-monotonic 46
- non-sequential check 392
- non-sequential hold check 393
- non-sequential setup check 393
- non-sequential timing check 365
- non-unate 34, 68
- N-well 417

- O**
- OCV 366
- OCV derating 371
- on-chip variation 365
- opening edge 377
- operating condition 39, 96, 472
- operating mode 418
- output current 79
- output external delay 257
- output fall 56
- output high drive 21
- output low drive 21
- output rise 56
- output specifications 206
- output switching power 88
- output_current_fall 79
- output_current_rise 80
- output_threshold_pct_fall 25
- output_threshold_pct_rise 26
- output_voltage_fall 83
- output_voltage_rise 83
- overshoot 87
- overshoot glitch 152, 159

- P**
- parallel PMOS 17
- parasitic corners 418
- parasitic extraction 532
- parasitic information 531
- parasitic interconnect 104
- parasitic RC 7
- path delay 34, 496
- path exception 444
- path groups 209
- path segmentation 224
- pathpulse delay 475
- pathpulsepercent delay 477
- paths 207
- PCB interconnect 349
- period 181, 513
- period timing check 483
- physical hierarchy 542
- physical net 532, 545
- physical partition 541, 547
- physical port 540
- physical wireload 547
- pie model 532, 548
- pi-model 104
- pin capacitance 20, 44, 537, 543
- pin-to-pin delay 470
- place-and-route 532
- PLL 10
- PMOS 15
- PMOS device 415
- PMOS transistor 16
- point-to-point delay 471
- port delay 477, 479, 517
- port slew 534
- positive crosstalk delay 166, 167
- positive fall delay 170
- positive glitch 150
- positive rise delay 170
- positive slack 247
- positive unate 33, 59
- positive_unate 56

post-layout phase 104
 power 12
 power definition 534, 540
 power dissipation 19
 power gating 414
 power gating cell 415
 power net 540
 power unit 100
 pre-layout phase 104
 process 534
 process operating condition 482
 process technology 12
 propagated_noise 158
 propagated_noise_high 83
 propagated_noise_low 83
 propagation delay 25, 477, 502
 pull-down structure 17
 pull-up resistance 21
 pull-up structure 17
 pulse propagation 469, 470
 pulse rejection limit 476, 481
 pulse width 476, 514
 pulse width check 66
 PVT 39, 336
 PVT condition 366, 371
 PVT corner 418
 P-well 417

Q

quarter-cycle delay 343

R

R value 534
 R_NET 532, 542
 RC 7, 103
 RC interconnect 103
 RC network 23, 544, 548
 RC time constant 23
 RC tree 108
 read cycle 343
 receiver pin capacitance 76
 receiver_capacitance1_fall 77, 78
 receiver_capacitance1_rise 78
 receiver_capacitance2_fall 77, 78
 receiver_capacitance2_rise 77, 78
 recovery 66
 recovery check 435
 recovery check arc 60
 recovery time 66, 511
 recovery timing check 279, 470, 483
 reduced format 115

reduced net 532, 542, 548
 reduced RC 545
 reduced representation 118
 reference_time 80
 related clocks 305
 related pin 385, 392
 related_pin 63
 removal 66
 removal check arc 60
 removal time 66, 512
 removal timing check 277, 470, 483
 resistance identifier 544
 resistance section 544, 547
 resistance unit 100, 538, 544
 resistive tree 103
 retain definition 477
 retain delay 478
 rise delay 51
 rise glitch 152, 159
 rise transition 51
 rise_constraint 64
 rise_glitch 154
 rise_transition 51
 rising_edge 69
 r-limit 476
 root-mean-squared 160
 routing confidence 536
 routing halo 177
 RSPF 113
 RTL 5

S

same-cycle checks 385, 389
 SBPF 113
 scan mode 65, 162
 scenarios 421
 SDC xvii, 4, 447
 SDC commands 447
 SDC file 447
 SDF xvi, 94, 418, 468
 sdf_cond 72, 95
 segment 101
 segmented wireload mode 110
 selection groups 113
 sequential arc 74
 sequential cell 33, 60
 series NMOS 17
 set 448
 set_case_analysis 219, 461
 set_clock_gating_check 395, 407, 412, 455

- set_clock_groups 455
- set_clock_latency 31, 188, 236, 456
- set_clock_sense 456
- set_clock_transition 186, 456
- set_clock_uncertainty 31, 186, 456
- set_data_check 385, 457
- set_disable_timing 219, 434, 457
- set_drive 210, 461
- set_driving_cell 210, 461
- set_false_path 38, 219, 272, 457
- set_fanout_load 462
- set_hierarchy_separator 448
- set_ideal_latency 458
- set_ideal_network 458
- set_ideal_transition 458
- set_input_delay 203, 239, 321, 369, 440, 458
- set_input_transition 210, 213, 234, 462
- set_level_shifter_strategy 466
- set_level_shifter_threshold 466
- set_load 211, 242, 462
- set_logic_dc 462
- set_logic_one 462
- set_logic_zero 463
- set_max_area 217, 463
- set_max_capacitance 215, 463
- set_max_delay 222, 459
- set_max_dynamic_power 466
- set_max_fanout 217, 463
- set_max_leakage_power 466
- set_max_time_borrow 459
- set_max_transition 215, 463
- set_min_capacitance 464
- set_min_delay 222, 459
- set_multicycle_path 219, 260, 460
- set_operating_conditions 41, 464
- set_output_delay 206, 257, 325, 369, 440, 460
- set_port_fanout_number 464
- set_propagated_clock 189, 461
- set_resistance 464
- set_timing_derate 368, 464
- set_units 448
- set_wire_load_min_block_size 465
- set_wire_load_mode 110, 465
- set_wire_load_model 465
- set_wire_load_selection_group 113, 465
- setup 62
- setup capture edge 315
- setup check 3, 227
- setup check arc 60
- setup constraint 63
- setup launch edge 251
- setup multicycle 289
- setup multicycle check 260
- setup receiving edge 251
- setup time 62, 508
- setup timing check 228, 470, 474, 482, 510
- setup_falling 393
- setup_rising 393
- setup_template_3x3 64
- shield wires 176
- shielding 177
- shortest path 35
- sidewall 148
- sidewall capacitance 118
- signal integrity 147
- signal traces 148
- simulation 467
- skew 30, 515
- sleep mode 414
- slew 28, 53
- slew derate factor 54
- slew derating 56
- slew rate 120
- slew threshold 54, 537
- slew_derate_from_library 55
- slew_lower_threshold_pct_fall 54
- slew_lower_threshold_pct_rise 54
- slew_upper_threshold_pct_fall 54
- slew_upper_threshold_pct_rise 54
- slow clock domain 289
- slow corner 229
- slow process 39, 96
- source latency 188
- source synchronous interface 317, 328
- specify block 496
- specify parameter 485
- specparam 474
- SPEF xvi, 113, 418, 531
- SPICE 113
- SRAM 317
- SRAM interface 336
- SSTA 427
- stage_type 82
- stamp event 482
- standard cell 19, 43
- standard delay annotation 467
- standard parasitic extraction format 531

standard Vt 416
standby mode 88
standby power 12, 88
startpoint 207
state-dependent model 70
state-dependent path delay 470
state-dependent table 59
statistical static timing analysis 427
statistical wireload 547
steiner tree 547
straight sum 169
subthreshold current 92
synchronous inputs 60
synchronous outputs 61
synthesis 471

T

temperature inversion 41
temperature variations 366
thermal budget 12
threshold specification 26
time borrowing 365, 379
time stamp 535
timescale 472
timing analysis 2, 532
timing arc 33, 45, 59, 94, 219, 392, 434
timing break 434
timing check 469, 470, 474, 482, 496
timing constraint 474
timing corner 370
timing environment 469, 471, 474, 485
timing model variable 474
timing paths 207
timing sense 33
timing simulation 2
timing specification 474
timing windows 161
timing_sense 51
timing_type 64, 69
T-model 103
top wireload mode 110
total capacitance 537, 542, 548
total power 416
transition delay 480
transition time 23, 28
triplet form 549
triplets 482, 534
TYP 41
typical delay 482
typical leakage 41

typical process 39, 96, 534
typical value 549

U

unateness 34
uncertainty 186
undershoot 87
undershoot glitch 152, 159
unidirectional skew timing
 check 483
units 99
upper metal layer 120
USB core 43
useful skew 444

V

valid endpoints 207
valid startpoints 207
Verilog HDL 4, 467, 485, 496, 536
version number 472, 534
VHDL 4, 467, 485, 499
VHDL87 536
VHDL93 536
vias 102
victim 149
victim net 150, 167
VIH 154
VIHmin 19
VIL 154
VILmax 19
virtual clock 217
virtual flip-flop 318
VITAL 499
VOH 154
VOL 154
voltage source 115
voltage threshold 366
voltage unit 100
voltage waveform 23

W

waveform specification 183
WCL 420
WCS 40, 420
well bias 417
when condition 71, 94
wide trace 120
width timing check 483
wildcard character 473
wire capacitance 148

INDEX

wireload mode 110
wireload model 7, 105
wireload selection group 112
worst-case cold 420
worst-case process 534
worst-case slow 40, 227, 370, 420
worst-case tree 108
worst-case value 549
write cycle 348

X

X filter limit 481
X handling 10

Z

zero delay 30
zero violation 246
zero-cycle checks 385
zero-cycle setup 389

